
VISUAL ANALYTICS: INTRO TO HTML, CSS, AND JAVASCRIPT

AB MOSCA (THEY/THEM)



TODAY'S PLAN

- Front end web development for visualization:
 - Modeling a webpage
 - Creating a webpage
 - Styling a webpage
 - Adding interaction to a webpage

MODELING A WEBPAGE

- Go to: smith.edu
- Right click anywhere, select “inspect” from the menu
- Start on the “Elements” pane of the inspection panel, explore the webpage
- How do you think this webpage is made?

MODELING A WEBPAGE

- A webpage is defined through HTML



HTML stands for **H**yper **T**ext **M**arkup **L**anguage

HTML is the **standard markup** language for Web pages

HTML **elements** are the building blocks of HTML pages

HTML elements are represented by **<>** tags

```
...<!DOCTYPE html> == $0
<html lang="en" dir="ltr" prefix="og: https://ogp.me/ns#" class=" js">
  > <head>...</head>
  > <body class="domain-smith_college" data-component-theme="neutral">...
    > <a href="#main-content" class="visually-hidden focusable"> Skip to
content </a>
    > <div class="dialog-off-canvas-main-canvas" data-off-canvas-main-ca
    > <div class="layout-container">
      > <div></div>
      > <div></div>
      > <div class="site-header__container">...</div>
      > <div class="main">
        > <a id="main-content" tabindex="-1"></a>
        > <main role="main" class="main-content">
          > <div data-swiftype-index="true">...</div>
          </main>
        </div>
        > <footer data-component-width="wide" class="site-footer">...</fo
      </div>
      > <style type="text/css">...</style>
      > <link rel="icon" type="image/png" sizes="32x32" href="https://www
ites/default/files/2025-02/favicon.png">
    </div>
```

https://www.w3schools.com/whatis/whatis_html.asp

MODELING A WEBPAGE

- DOM = Document Object Model
 - When a webpage is loaded, the browser (ex. Chrome) creates a DOM of the page
 - The DOM model of a webpage is a tree of HTML objects

HTML



The **HTML DOM** is an **Object Model** for **HTML**. It defines:

- HTML elements as **objects**
- **Properties** for all HTML elements
- **Methods** for all HTML elements
- **Events** for all HTML elements

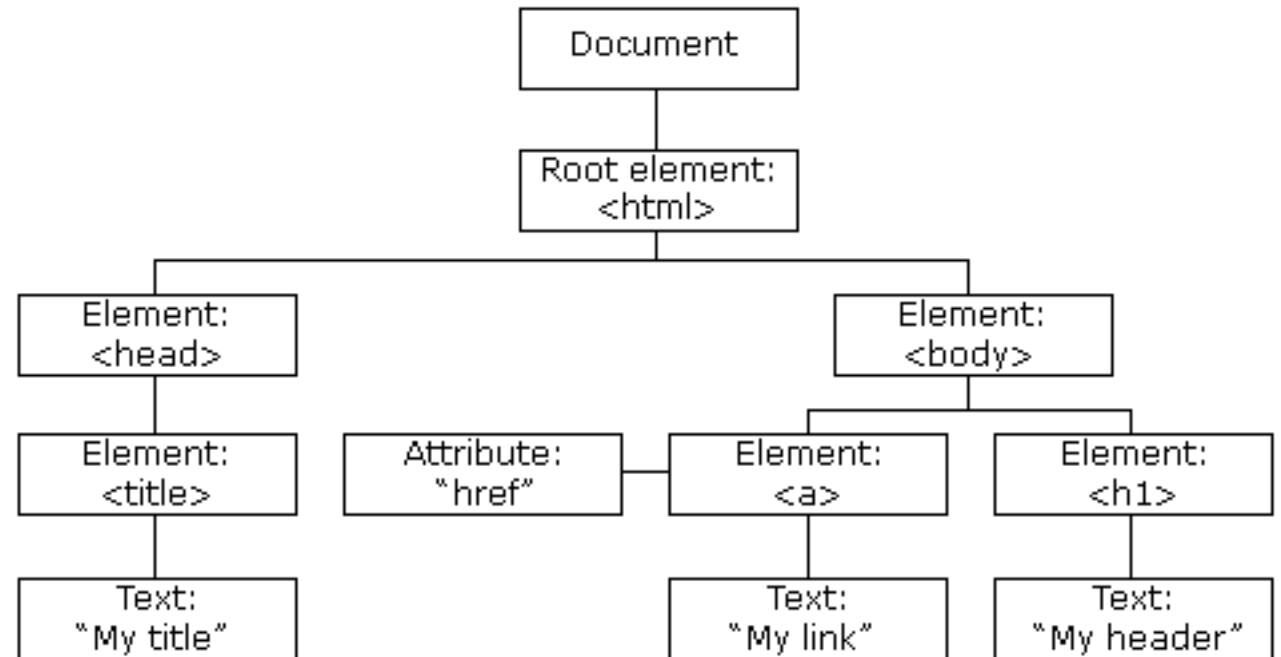


The **HTML DOM** is an **API** (Programming Interface) for **JavaScript**:

- JavaScript can add/change/remove HTML elements
- JavaScript can add/change/remove HTML attributes
- JavaScript can add/change/remove CSS styles
- JavaScript can react to HTML events
- JavaScript can add/change/remove HTML events

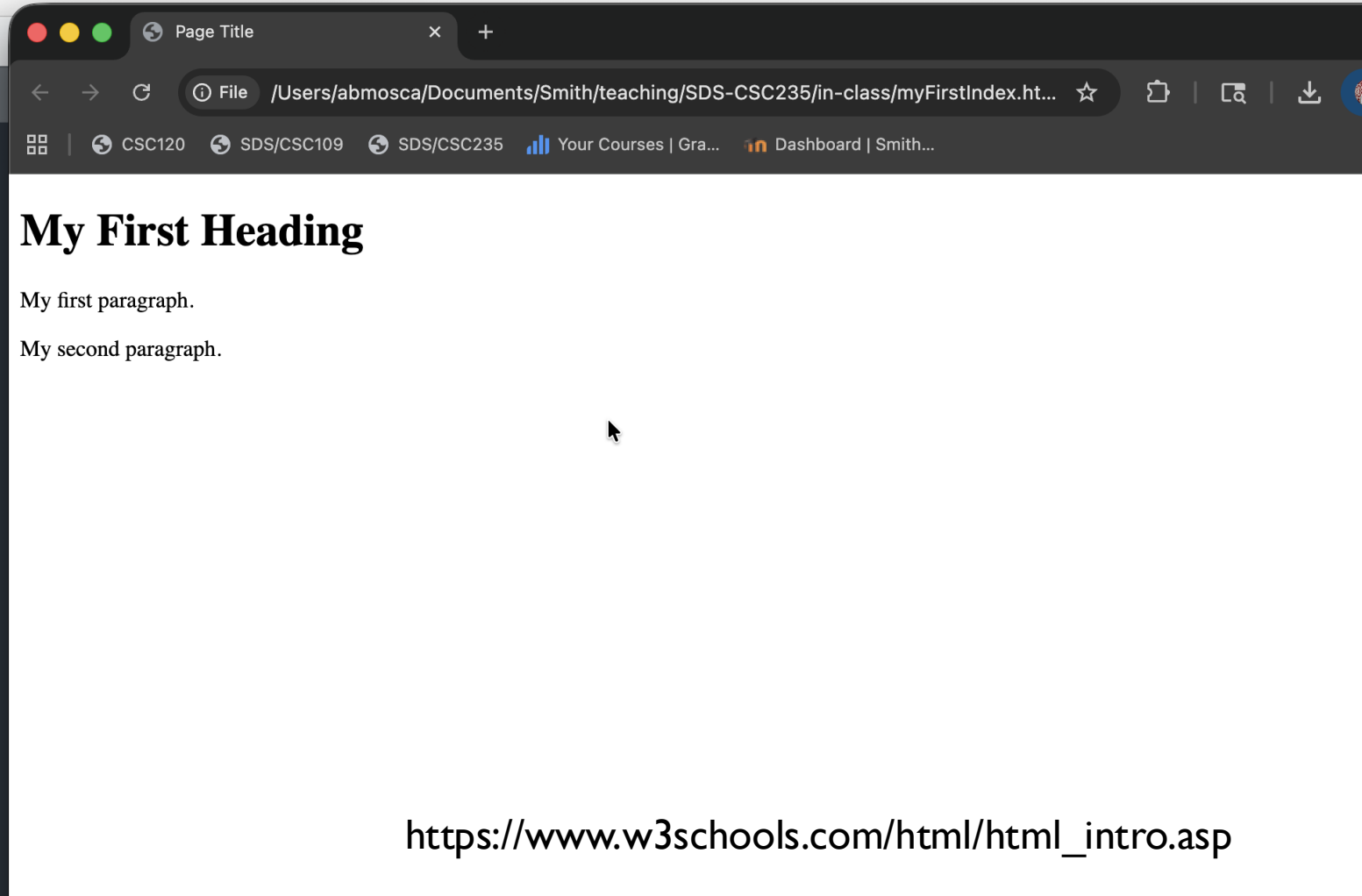
MODELING A WEBPAGE

- DOM = Document Object Model
 - When a webpage is loaded, the browser (ex. Chrome) creates a DOM of the page
 - The DOM model of a webpage is a tree of HTML objects



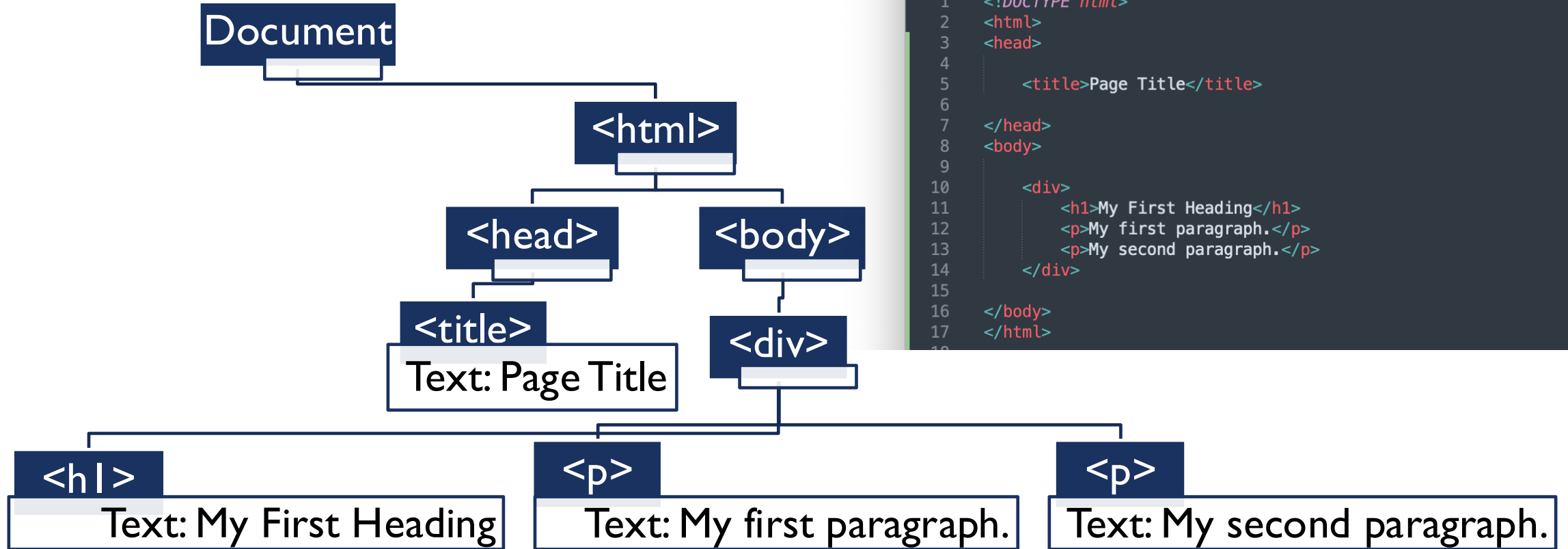
BUILDING A WEBPAGE: HTML

```
myFirstIndex.html
1  <!DOCTYPE html>
2  <html>
3  <head>
4
5      <title>Page Title</title>
6
7  </head>
8  <body>
9
10     <div>
11         <h1>My First Heading</h1>
12         <p>My first paragraph.</p>
13         <p>My second paragraph.</p>
14     </div>
15
16 </body>
17 </html>
18
19
```



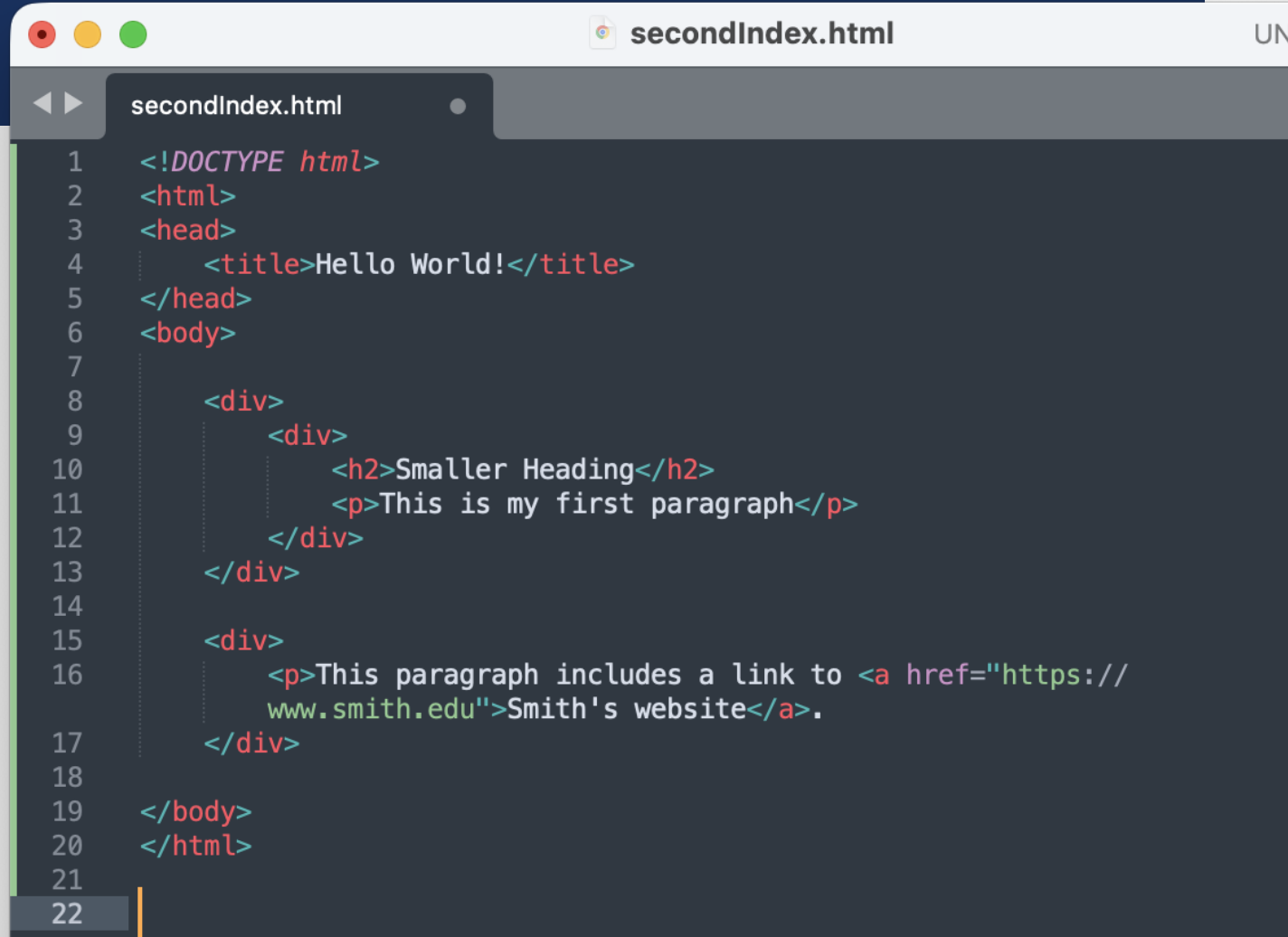
https://www.w3schools.com/html/html_intro.asp

BUILDING A WEBPAGE: HTML



BUILDING A WEBPAGE: HTML

Your turn: Draw the DOM for this webpage

A screenshot of a code editor window titled 'secondIndex.html'. The editor shows an HTML document with the following structure: a DOCTYPE declaration, an html root element, a head element containing a title 'Hello World!', and a body element. The body contains two div elements. The first div contains a smaller heading 'Smaller Heading' and a paragraph 'This is my first paragraph'. The second div contains a paragraph that includes a link to 'https://www.smith.edu' with the text 'Smith's website'. The code is color-coded: red for tags, green for text, and blue for the href attribute value. Line numbers 1 through 22 are visible on the left side of the editor.

```
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>Hello World!</title>
5  </head>
6  <body>
7
8      <div>
9          <div>
10             <h2>Smaller Heading</h2>
11             <p>This is my first paragraph</p>
12          </div>
13      </div>
14
15      <div>
16          <p>This paragraph includes a link to <a href="https://
17             www.smith.edu">Smith's website</a>.
18      </div>
19  </body>
20  </html>
21
22
```

BUILDING A WEBPAGE: HTML

- The `<!DOCTYPE html>` declaration defines that this document is an HTML5 document
- The `<html>` element is the root element of an HTML page
- The `<head>` element contains meta information about the HTML page
- The `<title>` element specifies a title for the HTML page (shown in the browser's title bar or in the page's tab)
- The `<body>` element defines the document's body, and is a container for all the visible contents
- The `<h1>` element defines a large heading
- The `<p>` element defines a paragraph
- Complete tag reference: <https://www.w3schools.com/tags/default.asp>

STYLING A WEBPAGE: CSS

- A webpage is styled through CSS

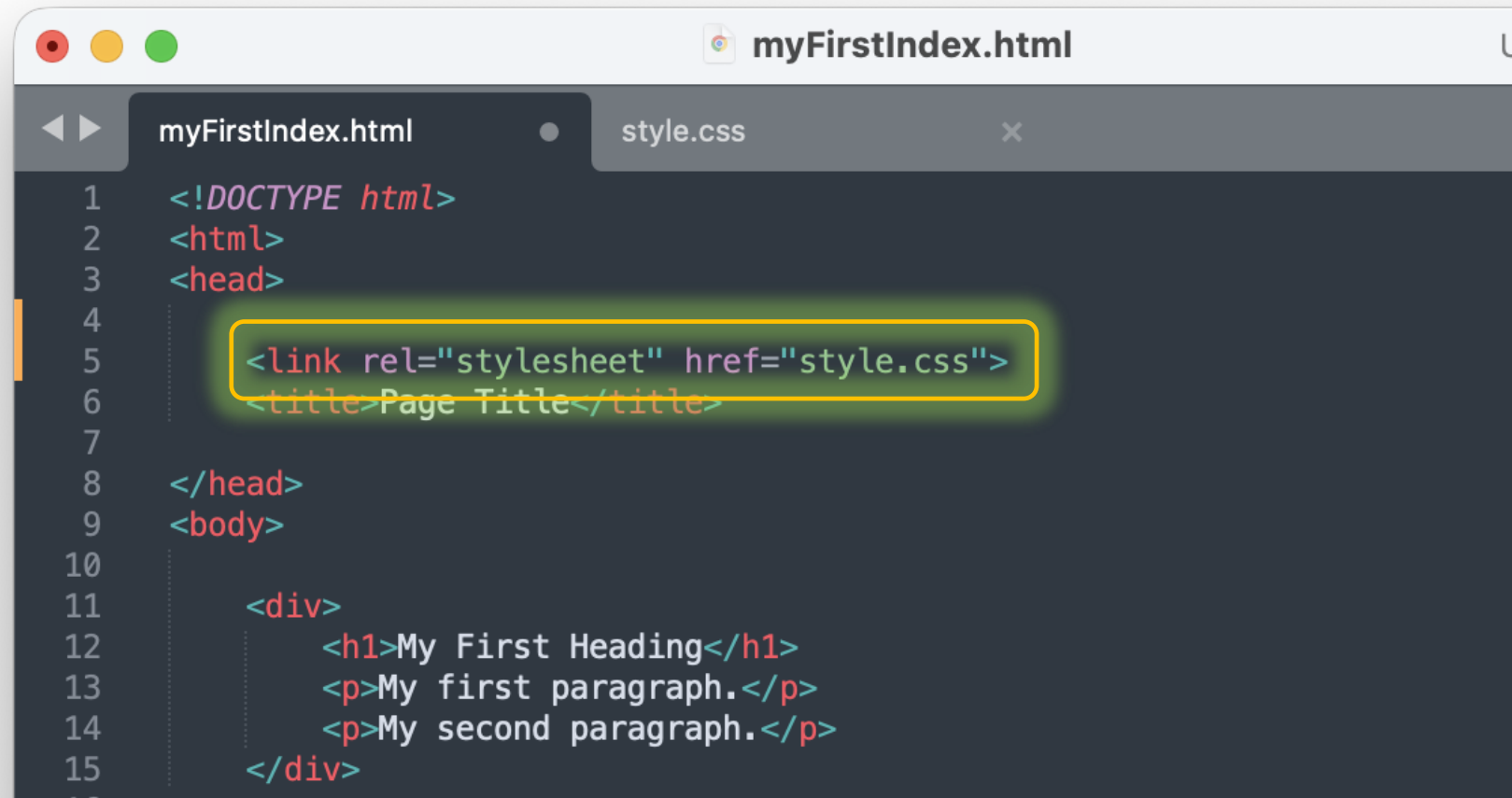


CSS stands for **C**ascading **S**tyle **S**heets

CSS describes how **HTML** elements are to be **displayed**

STYLING A WEBPAGE: CSS

- An HTML file is linked to a CSS file
- You can also style in-line and with internal CSS, but it is not best practice



The screenshot shows a code editor with two tabs: 'myFirstIndex.html' and 'style.css'. The 'myFirstIndex.html' tab is active, displaying the following HTML code:

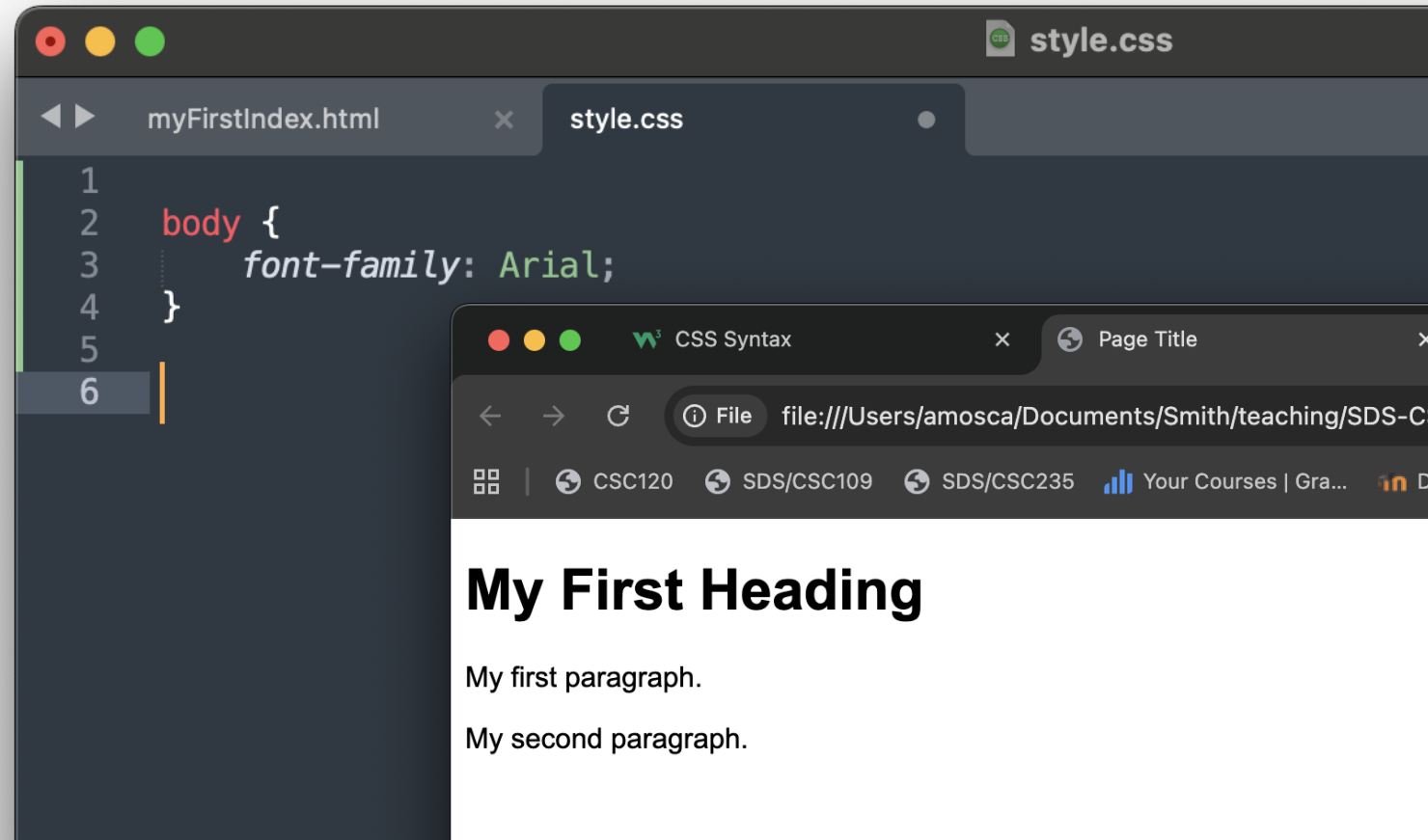
```
1  <!DOCTYPE html>
2  <html>
3  <head>
4
5  <link rel="stylesheet" href="style.css">
6  <title>Page Title</title>
7
8  </head>
9  <body>
10
11  <div>
12      <h1>My First Heading</h1>
13      <p>My first paragraph.</p>
14      <p>My second paragraph.</p>
15  </div>
```

The line containing the `<link rel="stylesheet" href="style.css">` tag is highlighted with a yellow box.

STYLING A WEBPAGE: CSS

- CSS files are sets of rules with specific syntax:

```
selector {  
    property : value;  
    property2 : value2;  
    ...  
}
```



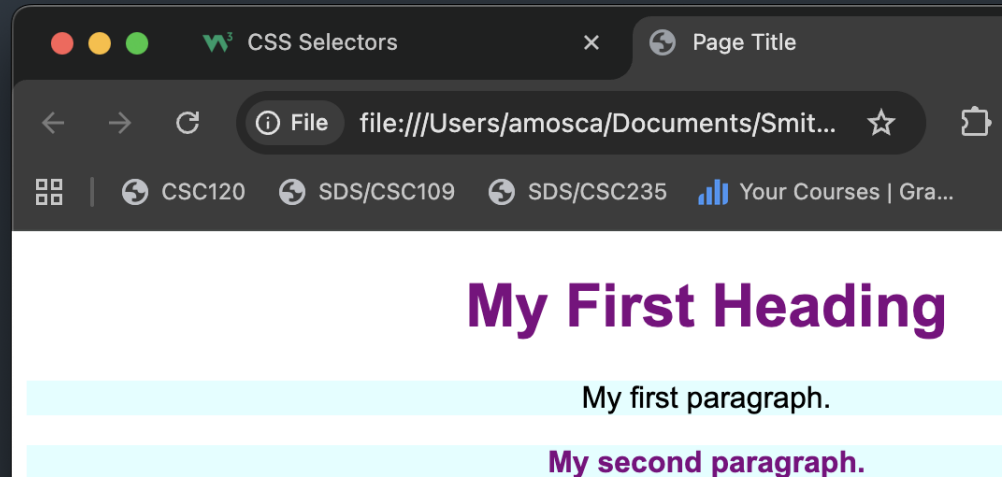
STYLING A WEBPAGE: CSS

- A **CSS selector** can be an
 - **element**
 - The selector `p` indicates style for all paragraphs
 - **class**
 - An HTML element can be assigned a class
 - Ex. `<h1 class="title">`
 - The selector `.title` indicates style for all elements with class "title"
 - **id**
 - An HTML element can be assigned an id
 - Ex. `<div id="landing">`
 - The selector `#landing` indicates style for the element with id "landing"

STYLING A WEBPAGE: CSS

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4
5     <link rel="stylesheet" href="style.css">
6     <title>Page Title</title>
7
8 </head>
9 <body>
10
11     <div id="landing">
12         <h1 class="important">My First Heading</h1>
13         <p>My first paragraph.</p>
14         <p class="important">My second paragraph.</p>
15     </div>
16
17 </body>
18 </html>
```

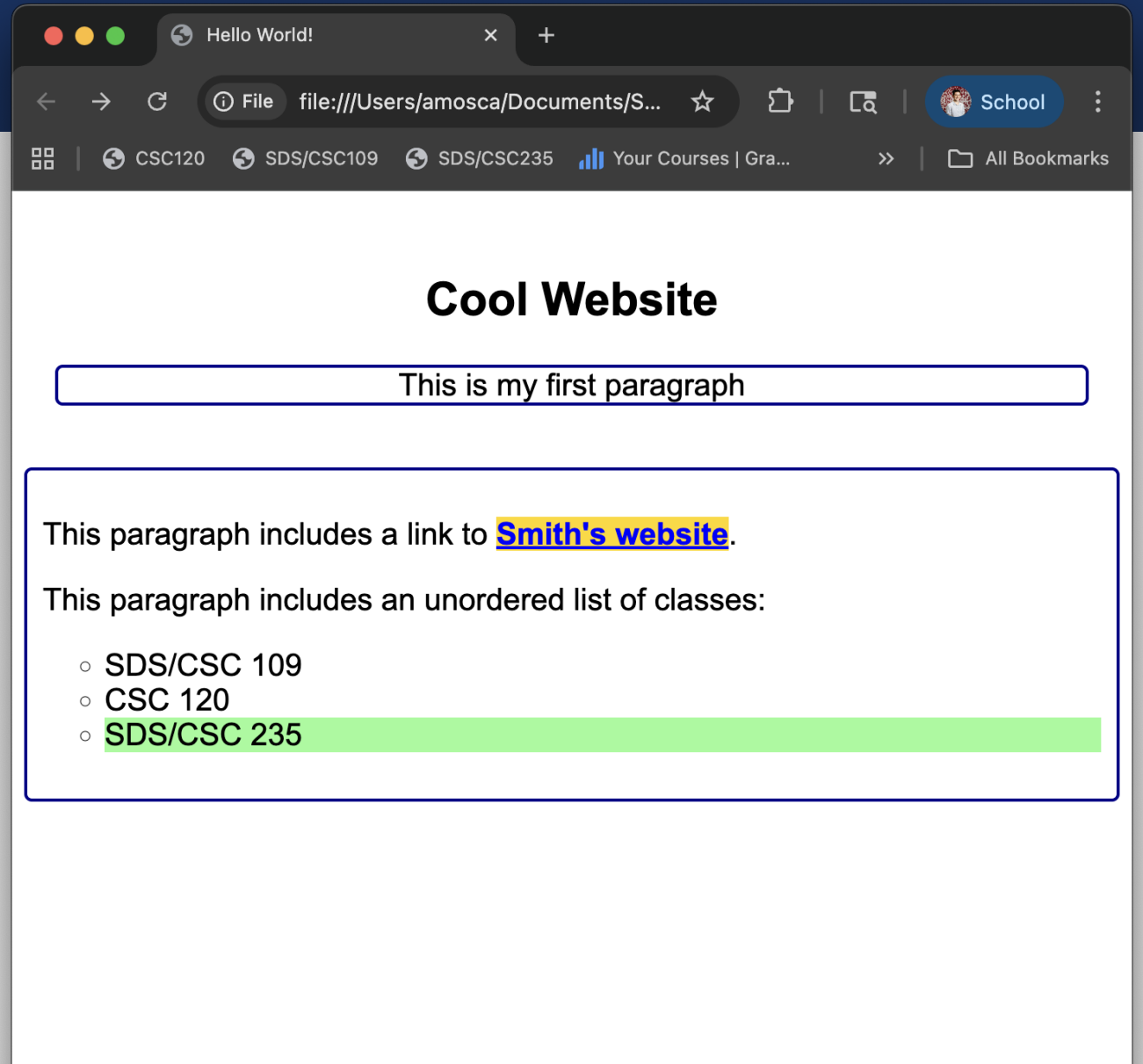
```
1
2 body {
3     font-family: Arial;
4 }
5
6 p {
7     background: lightcyan;
8 }
9
10 .important {
11     font-weight: bold;
12     color: purple;
13 }
14
15 #landing {
16     text-align: center;
17 }
```



STYLING A WEBPAGE: CSS

Your turn:

- Modify the HTML in secondIndex.html: link an external CSS file to create a webpage that looks like this →
- Helpful Resources:
 - <https://www.w3schools.com/html/>
 - <https://www.w3schools.com/css/>



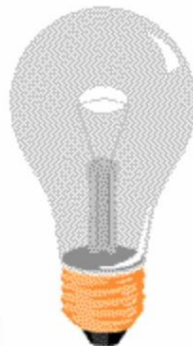
INTERACTIVE WEBPAGES: JS

- A webpage is made interactive through JavaScript (JS)
- JS can update and change HTML and CSS

Light bulb example

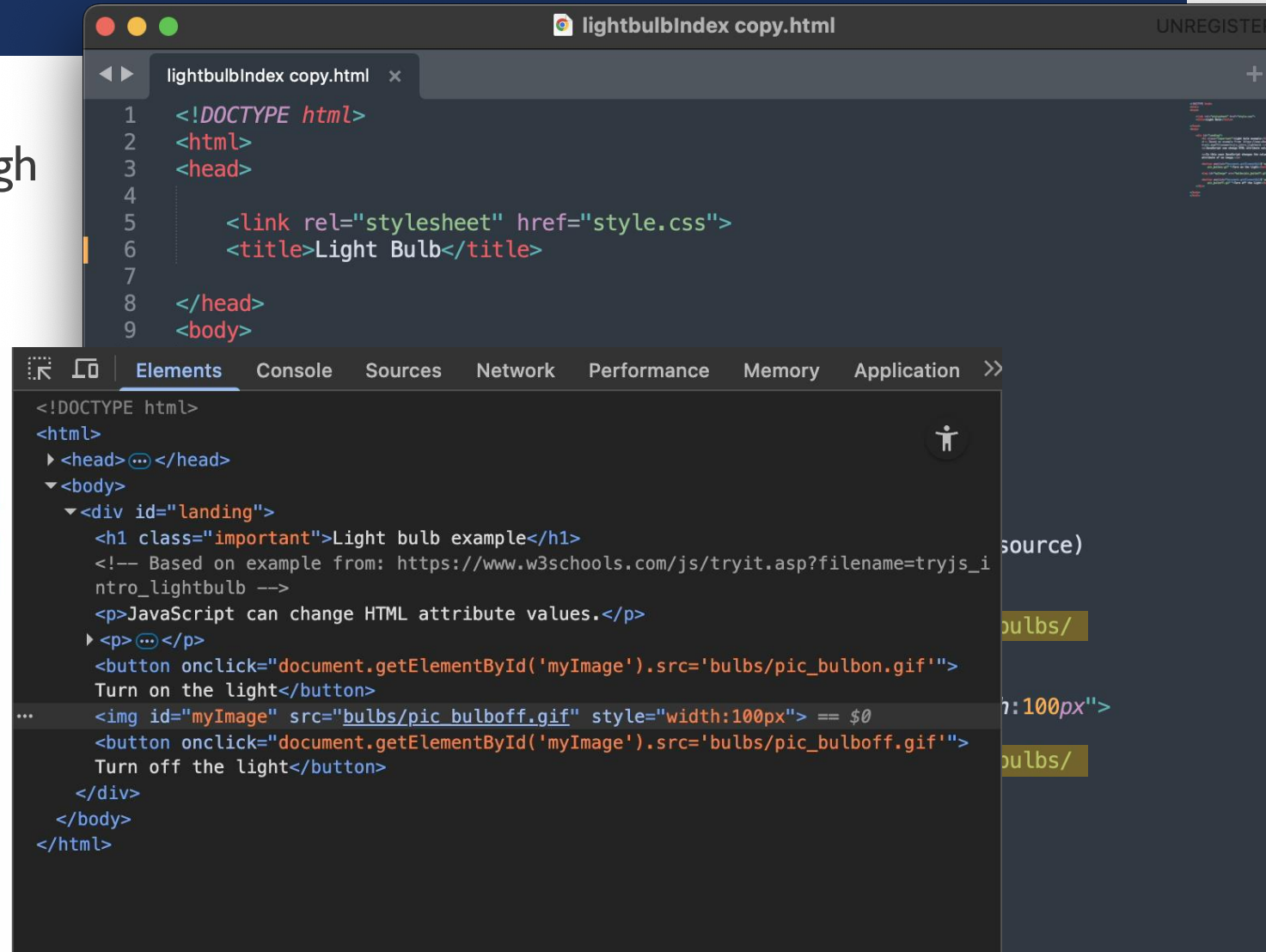
JavaScript can change HTML attribute values.

In this case JavaScript changes the value of the src (source) attribute of an image.



Turn on the light

Turn off the light

A screenshot of a web browser window titled 'lightbulbIndex copy.html'. The browser shows a web page with a glowing light bulb and two buttons: 'Turn on the light' and 'Turn off the light'. The 'Elements' panel is open, showing the HTML structure. The code in the source view is as follows:

```
<!DOCTYPE html>
<html>
<head>
  <link rel="stylesheet" href="style.css">
  <title>Light Bulb</title>
</head>
<body>
  <div id="landing">
    <h1 class="important">Light bulb example</h1>
    <!-- Based on example from: https://www.w3schools.com/js/tryit.asp?filename=tryjs_intro_lightbulb -->
    <p>JavaScript can change HTML attribute values.</p>
    <p><button onclick="document.getElementById('myImage').src='bulbs/pic_bulbon.gif'">
      Turn on the light</button>
    
    <button onclick="document.getElementById('myImage').src='bulbs/pic_bulboff.gif'">
      Turn off the light</button>
  </div>
</body>
</html>
```

INTERACTIVE WEBPAGES: JS

- Syntax:
 - Literals: numbers, strings
 - Ex. `10.0`, `10`, `"ten"`, `'10'`
 - Declaring variables:
 - `let x = 4;`
 - `const PI = 3.14;`
 - Typically, lowerCamelCase is used
- Semicolons separate JS statements
- JS ignores whitespace
- `//` for single line comments
- `/*` for multi-line comments `*/`

INTERACTIVE WEBPAGES: JS

- HTML events are “things” that happen to HTML elements
 - Ex. Webpage finished loading, input field was changed, button was clicked
- JavaScript can “react” on these events
- How?
 - Assign an HTML element an event handler attribute with JS code

```
17
18  <button onclick="document.getElementById('myImage').src='bulbs/
19    pic_bulbon.gif'">Turn on the light</button>
20
21  
```

INTERACTIVE WEBPAGES: JS

- HTML events:
 - onchange
 - onclick
 - onmouseover
 - onmouseout
 - onkeydown
 - onload
 - More: https://www.w3schools.com/jsref/dom_obj_event.asp

INTERACTIVE WEBPAGES: JS

- JS + HTML
 - JS can be included as an HTML attribute Ex. `onclick="doSomething();"`
 - JS can be included between `<script></script>` tags
 - You can have multiple
 - They can be in the `<body>`, `<head>`, or both
 - You can write JS between the `<script>` tags in the .html file **OR** use an external file with the `src` attribute

INTERACTIVE WEBPAGES: JS

```
betterLightbulbIndex.html x
1 <!DOCTYPE html>
2 <html>
3 <head>
4
5 <link rel="stylesheet" href="style.css">
6 <title>Light Bulb</title>
7
8 </head>
9 <body>
10
11 <div id="landing">
12 <h1 class="important">Better Light bulb example</h1>
13 <!-- Based on example from: https://www.w3schools.com/js/
14 tryit.asp?filename=tryjs_intro_lightbulb -->
15 <p>JavaScript can change HTML attribute values.</p>
16
17 <p>In this case JavaScript changes the value of the src (source)
18 attribute of an image.</p>
19
20 <button onclick="lightOn()">Turn on the light</button>
21
22 
23
24 <button onclick="lightOff()">Turn off the light</button>
25
26 </div>
27
28 <script src="index.js"></script>
29
30 </body>
31 </html>
```

```
index.js
1
2 // Event handler for turn light on button
3 function lightOn() {
4     document.getElementById('myImage').src='bulbs/pic_bulbon.gif';
5 }
6
7 // Event handler for turn light off button
8 function lightOff() {
9     document.getElementById('myImage').src='bulbs/pic_bulboff.gif';
10 }
```

Better Light bulb example

JavaScript can change HTML attribute values.

In this case JavaScript changes the value of the src (source) attribute of an image.



Turn on the light

Turn off the light

INTERACTIVE WEBPAGES: JS

Your turn:

- Modify betterLightbulbIndex.html, myStyle.css, and index.js as needed to create a webpage that looks and functions like this →
- Helpful Resources:
 - <https://www.w3schools.com/html/>
 - <https://www.w3schools.com/css/>
 - <https://www.w3schools.com/js/>

Better Light bulb example

JavaScript can change HTML attribute values.

In this case JavaScript changes the value of the src (source) attribute of an image.



Turn on the light

Turn off the light